Smart Exam Hall Monitoring and Management System

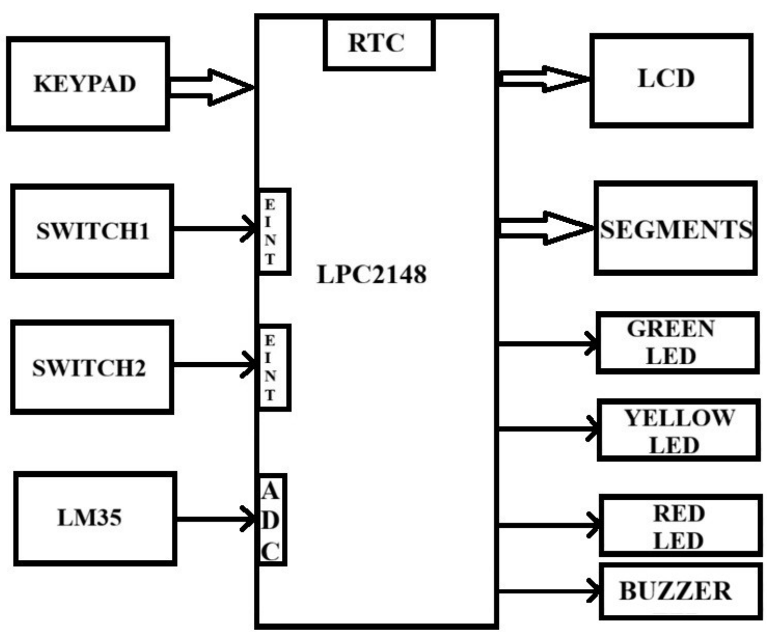
AIM:

To develop a Smart Exam Hall Monitoring System that automates examination timing, environmental monitoring, and status indication, thereby improving examination management efficiency and reducing manual intervention.

Objectives:

1. To display the current RTC time and room temperature on the LCD.
2. To allow examination duration configuration using a keypad.
3. To implement a countdown timer for examination management.
4. To display the remaining examination time using two multiplexed 7-segment displays.
5. To provide Green, Yellow, and Red LED alerts based on the remaining examination time.
6. To enable password-protected RTC editing, examination start-time configuration, and examination duration configuration using External Interrupt 0.
7. To implement Pause/Resume functionality using External Interrupt 1.
8. To continuously monitor and display room temperature using an LM35 sensor.
9. To automatically record examination start and end times using the RTC.
10. To reduce manual timing errors and improve examination hall monitoring efficiency.

BLOCK DIAGRAM:



REQUIREMENTS:

HARDWARE REQUIREMENTS:

- LPC2148
- 16X2 LCD
- 4X4 MATRIX KEYPAD
- SWITCHES
- SEVEN SEGMENTS
- LM35
- LED'S
- BUZZER
- USB-UART CONVERTER / DB-9 CABLE

SOFTWARE REQUIREMENTS:

- EMBEDDED-C PROGRAMMING
- FLASH MAGIC

PROJECT WORKFLOW:

The Smart Exam Hall Monitoring and Management System is an intelligent embedded solution designed to improve the efficiency, accuracy, and monitoring capabilities of examination halls. The system integrates an RTC (Real-Time Clock), LCD, two multiplexed 7-segment displays, keypad, LM35 temperature sensor, LEDs, and external interrupt mechanisms to provide automated examination timing and environmental monitoring.

Under normal operation, before the examination begins, the LCD continuously displays the current date and time obtained from the RTC along with the room temperature, allowing invigilators to monitor both timing and environmental conditions in real time. Using Switch-1 (External Interrupt 0), the invigilator can enter the RTC configuration mode to update the current date and time and configure the examination duration through the keypad. Using Switch-1 (External Interrupt 0), the invigilator can enter the secured configuration mode. Before allowing any modification, the system asks for an admin password through the 4x4 keypad. If the entered password is correct, the system allows the invigilator to edit the RTC date and time, set the examination start time, and configure the examination duration. If the password is wrong, access is denied and the system returns to normal monitoring mode.

Once the examination is started, the system records the exact start time using the RTC and initiates a countdown timer. During the examination, the LCD displays the current RTC time, room temperature, and remaining examination time, while the two multiplexed 7-segment displays prominently show the remaining examination time in minutes (00–99), enabling students and invigilators to easily track the progress of the examination.

To indicate examination status, a three-level visual warning system is implemented. The Green LED remains ON when more than 10 minutes are available. The Yellow LED activates during the final 10 minutes of the examination, while the Red LED turns ON during the last 1 minute, providing a clear visual indication that the examination is nearing completion. When the countdown reaches zero, the buzzer is activated to indicate the end of the examination session.

The system continuously monitors the room temperature using an LM35 sensor and displays the measured temperature on the LCD. This feature helps maintain a comfortable examination environment and allows invigilators to identify unfavourable temperature conditions that may affect student concentration and performance.

For handling interruptions, Switch-2 (External Interrupt 1) provides a pause/resume control for the countdown timer. Pressing the switch once pauses the countdown operation, while pressing it again resumes the timer from the exact point at which it was stopped. This feature ensures fair time management during unexpected disturbances, technical issues, or official announcements.

A unique feature of the proposed system is its ability to automatically log the examination start and end times using the RTC. These records can be displayed on the LCD and used for administrative verification and documentation purposes. By combining RTC-based timekeeping, countdown management, temperature monitoring, interrupt-driven control, and automatic event logging, the system provides a reliable, user-friendly, and cost-effective solution for modern examination halls.

**********************ALL THE BEST**********************

Note:

The application code must be developed following industry-standard embedded programming practices, with proper structure, naming conventions, modular design, and complete test case validation support. Any code not adhering to these standards will not be accepted.